

Elastic contact forces and independent angular couples

Verified sphere–plane examples

Rigid Body Collisions research prototype

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The model is a synthetic constitutive hypothesis, not a calibrated rubber law or a new friction law. It verifies an independent collision couple, explicit elastic storage, yielding and conditional reversal. Frozen source: 78e63313b5d1. 5 of 6 cases satisfy the registered gates; 15 histories and 3 rejected attempts are retained. Numerical timings were collected concurrently with native friction research and cannot establish a general speed ranking. This follow-up tightens integration tolerances only, with unchanged physics and gate budgets. The original four qualified cases, all six failures and two rejected attempts remain archived. Five additional cases now qualify, making nine of ten distinct examples verified across the studies. Weighted high-spin/low-friction remains rejected. Figures retain previously qualified original tangential/oblique traces and use refined traces otherwise.

A point-like wrench includes a couple

For a sphere of mass m , radius R and inertia $\mathbb{I} = \frac{2}{5}mR^2$, a fixed plane has normal $\hat{n}$ directed into the admissible half-space. The reference lever is $r = -R\hat{n}$; compression is $\delta = \max(0, R - \hat{n} \cdot x + d)$.

$$r\wedge = \begin{bmatrix} 0 & -r_z & r_y \\ r_z & 0 & -r_x \\ -r_y & r_x & 0 \end{bmatrix}, \quad v_c = v + (r\wedge)^\top \omega.$$

$$\Delta p = \int F dt, \quad \Delta L = \int \tau_n \hat{n} dt, \quad m\Delta v = \Delta p + \int mg dt, \quad \mathbb{I}\Delta\omega = (r\wedge)\Delta p + \Delta L.$$

The normal-axis spin change requires the independent couple ΔL : the offset point force has zero torque component along $\hat{n}$. A point-like wrench is a representation of effective contact mechanics, not a claim that a mathematical point traction carries a moment.

Elastic history, shared yield and energy

Let q be the two-component tangential displacement, θ the relative twist angle and a an effective material contact length. Define $h = (q_1, q_2, a\theta)^\top$, $K = \text{diag}(k_t, k_t, k_\theta/a^2)$ and $f = (\delta/R)^p$, with $p = 2$ for the default compression-weighted contact.

$$U = \frac{1}{2}k_n\delta^2 + \frac{1}{2}fh^\top Kh, \quad U_h = \frac{1}{2}fh^\top Kh, \quad F_n = k_n\delta + c_n \max(0, \dot{\delta}) + pU_h/\delta.$$

$$F_t = -fk_t q, \quad \tau_n = -fk_\theta \theta, \quad \sqrt{\|F_t\|^2 + (\tau_n/a)^2} \leq \mu k_n \delta \leq \mu F_n.$$

The last bound shares the normal-load budget between sliding and twisting. It is a phenomenological ellipsoidal capacity approximation, not an exact patch traction result. Using the elastic base normal load makes the budget conservative even when history or damping adds normal force. One coefficient μ covers sticking and sliding; this prototype does not yet fit distinct static and dynamic coefficients. Damping acts only during compression.

Let T contain orthonormal plane tangents and $u = (T^\top v_c, a\hat{n} \cdot \omega)^\top$. During elastic stick, $\dot{h} = u$. At the yield boundary, associated plastic flow has $\dot{h} = u - \lambda s$, with $s = fKh/\|fKh\|$ and $\lambda \geq 0$ chosen to prevent outward motion of the yield function. Consequently

$$\dot{D} = c_n \max(0, \dot{\delta})\dot{\delta} + \lambda\|fKh\| \geq 0, \quad \frac{d}{dt} \left(\frac{1}{2}m\|v\|^2 + \frac{1}{2}\mathbb{I}\|\omega\|^2 - mg \cdot x + U + D \right) = 0.$$

The default store vanishes smoothly at separation. The optional constant-stiffness benchmark uses $p = 0$ and explicitly records any residual history store as separation loss; matched oscillator tests return it to zero. No elastic state is silently deleted. Adaptive integration rejects an exhausted RHS evaluation budget.

Independent exact oscillators

For the linear benchmark $p = 0$, no damping and no yielding, choose

$$\omega_c = \sqrt{k_n/m}, \quad t_c = \pi/\omega_c, \quad k_\theta = \mathbb{I}k_n/m, \quad k_t = \frac{\mathbb{I}k_n}{\mathbb{I} + mR^2}.$$

The normal and twist modes then share a half period: $v_n^- = -V \rightarrow v_n^+ = V$, $\Omega_n^+ = -\Omega_n^-$. Throughout contact, $|\tau_n|/F_n = \mathbb{I}|\Omega_n^-|/(mV)$; the no-yield requirement is therefore $\mu a \geq \mathbb{I}|\Omega_n^-|/(mV)$, not merely an endpoint impulse inequality.

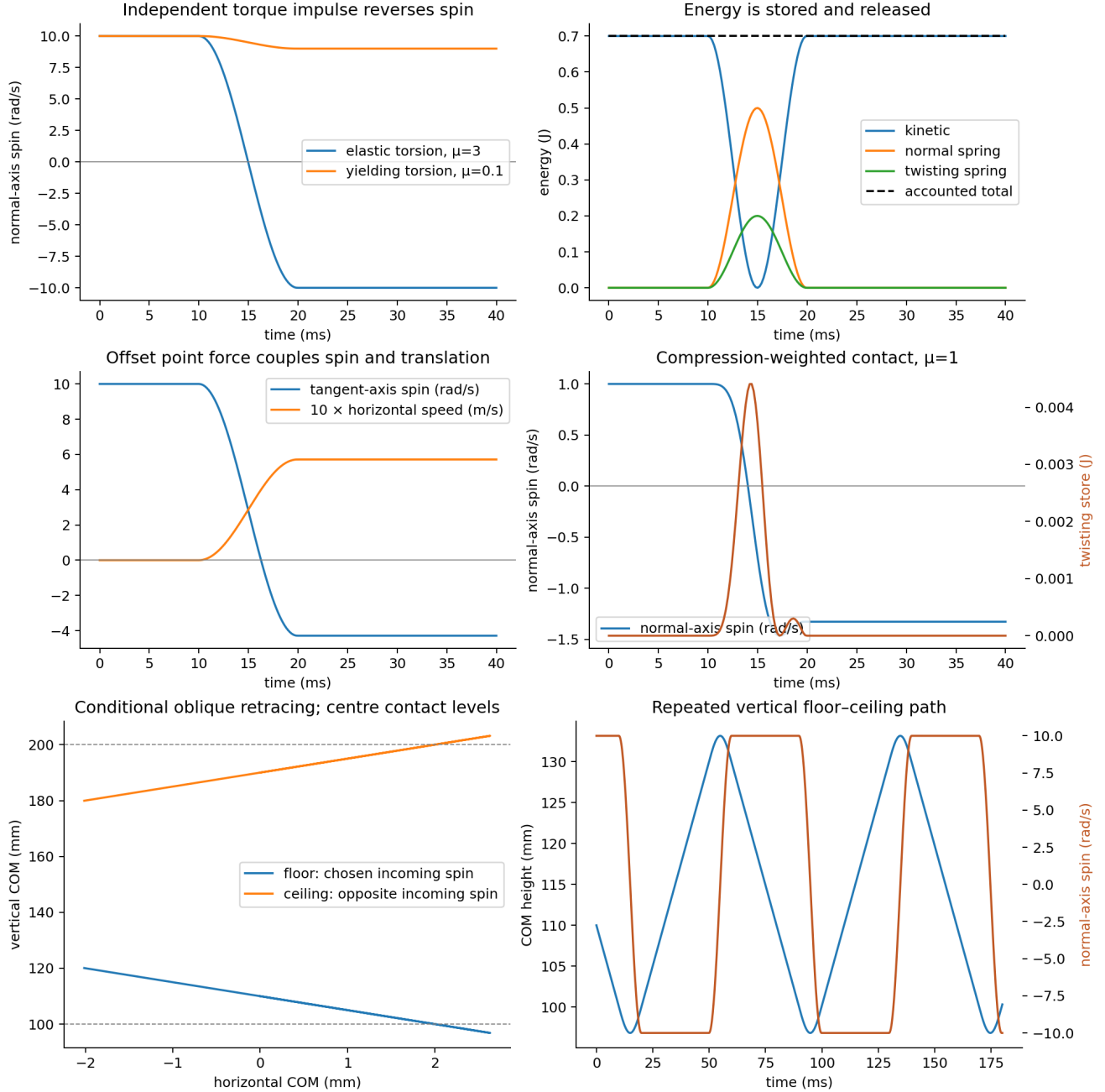
For a solid sphere dropped with zero horizontal COM speed and tangent-axis spin Ω , the same elastic tangential oscillator gives

$$\Omega_t^+ = -\frac{3}{7}\Omega_t^-, \quad v_t^+ = \frac{4}{7}R\Omega_t^-.$$

With $v_t^- = v_0$ and appropriately oriented $\Omega_t^- = -v_0/(\alpha R)$ at the floor, $\alpha = \mathbb{I}/(mR^2) = 2/5$, both horizontal velocity and spin reverse. A ceiling needs the opposite incoming spin. Arbitrary rubber-ball throws do not necessarily retrace. A vertical floor–ceiling path with pure normal-axis spin can repeat these exact elastic reversals.

Evidence and limitations

Case	Gates	Fine outgoing spin (rad/s)	Energy error (J)
elastic normal axis spin	yes	0, 0, -10	2.9e-14
vertical floor ceiling	yes	0, 0, -10	3.4e-14
low friction no spin reversal	yes	0, 0, 9	2.8e-14
compression weighted reversal	yes	0, 0, -1.3268	1.6e-14
rapid same material	yes	0, 0, -1000	3.9e-11
weighted high spin low friction budget	no	rejected	—



The weighted contact can transfer energy between normal and rotational channels, so a single-channel restitution may exceed one while total mechanics remain passive. Low friction can dissipate enough elastic input to prevent reversal. The same-material slow and rapid examples use speeds 0.01 and 100 m/s, demonstrating numerical behavior of this ideal law, not real rubber at those rates. All traces, failures, frozen parameters, source and three-level refinements are archived. This is a sphere/plane prototype; arbitrary-shape simultaneous contact needs separate engine validation. The accompanying literature review distinguishes known soft-finger mechanics from any unproven publication novelty.